

SALES INTELLIGENCE PROJECT

Sales Performance, Profitability and Business Intelligence Report

Total Sales \$25.66M	Gross Profit \$8.34M	Gross Margin 32.52%
Orders 10,889	Units 21,078	Average Order Value \$2.36K

This report documents the Sales Intelligence project built from the supplied retail-sales workbook. The project prepares a star-schema-oriented dataset, defines reusable business measures, answers management questions, and provides an interactive seven-page dashboard replica. The Power BI kit in the repository contains Power Query, DAX, theme, and build-guide assets for recreating the interactive report in Power BI Desktop.

Data period: 01 Jan 2021 to 15 Mar 2023 | Transactions: 10,889 | Products: 101 | Customers: 801 | Salespersons: 45 | Locations: 74

Prepared as a portfolio / academic project report

1. Executive Summary

The project consolidates 10,889 orders into a transaction-level Fact_Sales table and supporting dimensions. Across the available period, the dataset contains approximately \$25.66M in sales and \$8.34M in gross profit, producing a 32.52% gross margin. The average order value is approximately \$2.36K.

2021 generated \$11.69M and 2022 generated \$11.57M. 2023 is partial, covering only through 15 Mar 2023; its like-for-like year-to-date sales are \$2.40M, 3.7% versus the same period in 2022.

Project Objectives

- Measure revenue, orders, quantity, profitability, margin and average order value.
- Analyze daily, monthly, quarterly and yearly sales patterns.
- Identify high- and low-performing products, customers, salespersons and locations.
- Compare actual sales with the available budget structure while documenting data-quality assumptions.
- Provide interactive filtering and visual analysis for management-oriented questions.

Data Preparation and Model

The three yearly sales tables are consolidated into Fact_Sales. Dimension-oriented tables cover date, product, customer, salesperson and location analysis. The model follows a star-schema-oriented structure in which the transaction fact is filtered by descriptive dimensions.

Fact	Dimensions
Fact_Sales	Dim_Date; Dim_Product; Dim_Customer; Dim_Salesperson; Dim_Location
Fact_Budget	Dim_Date; Dim_Location

2. Sales Trends and Product Analysis

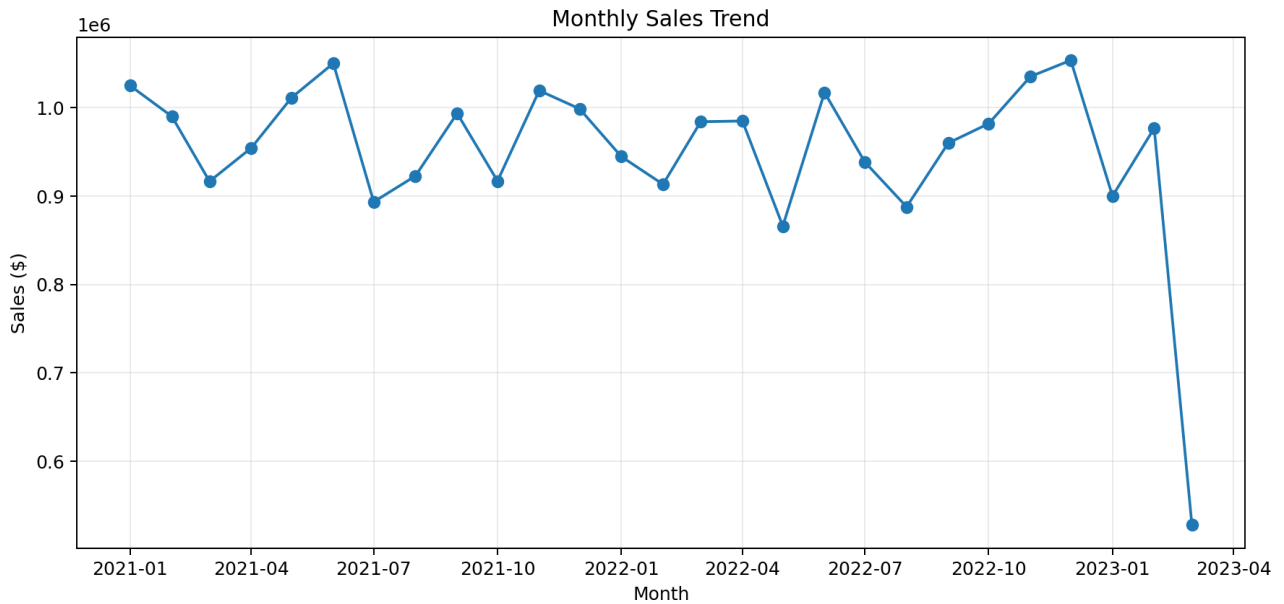


Figure 1. Monthly sales trend generated from the project dataset.

The trend view supports monthly performance analysis and provides the foundation for year-over-year and time-intelligence measures. The repository's DAX kit includes previous-year, MTD and YTD calculations.

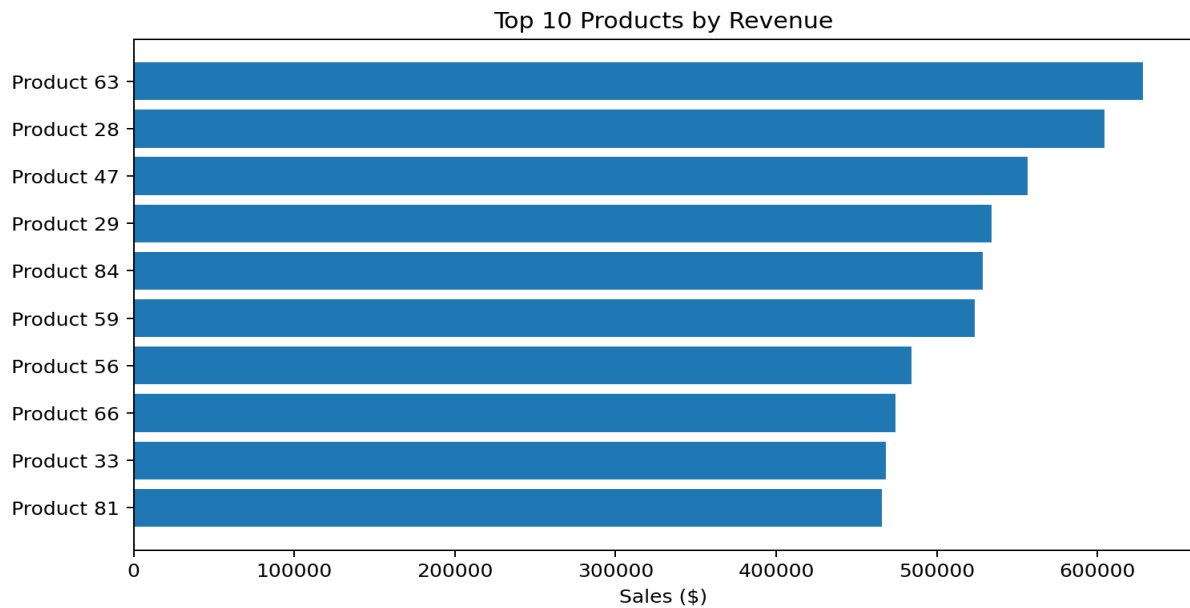


Figure 2. Top 10 products by revenue.

The product analysis shows meaningful variation in revenue contribution. The management-answer file identifies Product 63 as the highest-revenue product at approximately \$628K, followed by Product 28 at approximately \$604K.

3. Customer, Sales Team and Geographic Analysis

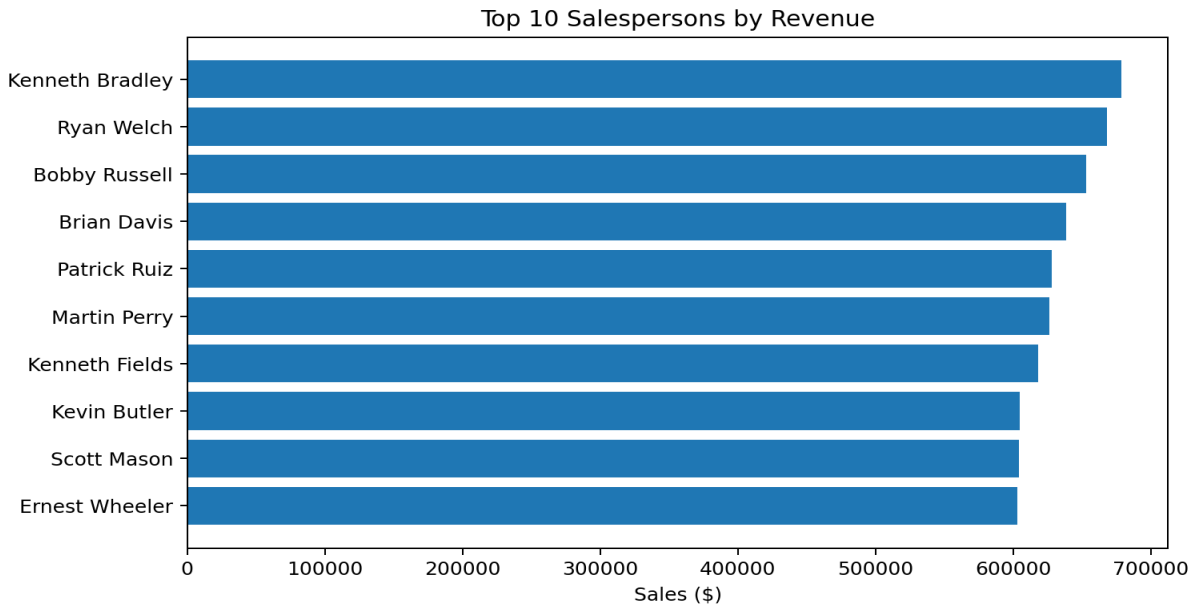


Figure 3. Top 10 salespersons by revenue.

The sales-team analysis compares revenue, profit and order volume. Kenneth Bradley is the highest-revenue salesperson in the generated analysis, while Ryan Welch has the highest order count.

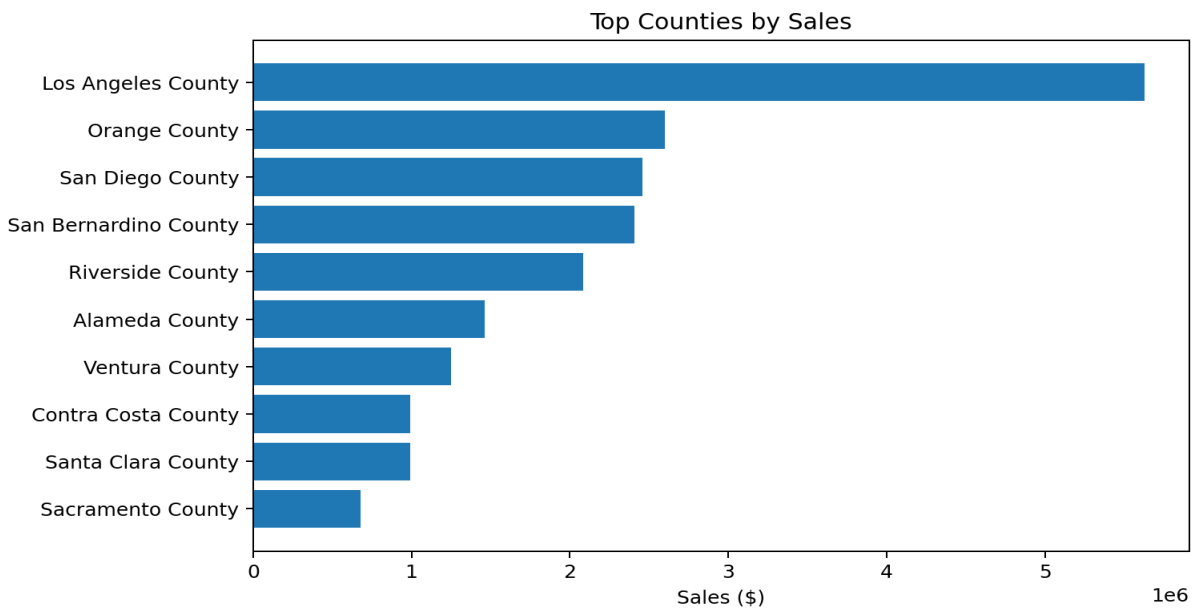


Figure 4. Leading counties by sales.

The location analysis is constrained to California at state level, so county and city views are more informative. Los Angeles County is the largest county contributor in the supplied dataset.

4. Profitability, Budget and Analytical Findings



Figure 5. Product sales versus gross profit.

Metric	Result
Total sales	\$25,661,209
Gross profit	\$8,343,893
Gross margin	32.52%
2021 sales	\$11,690,763
2022 sales	\$11,565,545
2023 YTD vs 2022 same window	+3.7%
Budget achievement (source-data assumption)	338%

Budget interpretation requires caution. The project guide documents that budget location IDs/names do not align cleanly with the location master, the budget has no explicit year, and 23 budget IDs do not match the available location set. The repository therefore treats the budget result as an assumption rather than a confirmed business KPI.

Key Findings

- Reported sales are approximately \$25.66M with a 32.5% gross margin.
- 2022 sales were slightly below 2021, while 2023 shows a positive like-for-like YTD comparison but is not a full year.
- Product contribution is uneven; the top products account for a meaningful share of revenue.
- Customer revenue is broadly distributed across the available customer IDs.
- Location analysis is more useful at county/city level because the dataset contains only California at state level.

5. Recommendations, Limitations and Conclusion

Recommendations

- Monitor high-sales products with relatively low margins and review pricing or cost structure.
- Prioritize high-profit products and investigate lower-performing products using both sales and margin.
- Use sales, profit and order volume together when reviewing salesperson performance.
- Focus geographic analysis on county/city performance rather than state-to-state comparisons.
- Rebuild or validate the budget mapping before using achievement percentages as a direct management KPI.

Important Limitations

- 2023 data is partial and ends on 15 March 2023.
- Only California is represented in the location state field.
- Customer names are not unique; Customer ID / CustomerLabel should be used for customer-level analysis.
- Budget periods and location mappings require validation before interpreting budget achievement as a business result.
- The current repository contains a Python/Plotly interactive replica plus Power BI build assets; the actual interactive Power BI .pbix file must be created in Power BI Desktop if the assignment requires a native PBIX deliverable.

Conclusion

The Sales Intelligence project turns the supplied transaction data into a structured analytical workflow using data preparation, dimensional modeling, reusable measures and interactive reporting. The report provides a clear portfolio-style record of the analysis while preserving the documented assumptions and limitations of the source data.

Repository contents

Folder/File	Purpose
src/	Python ETL, measures and dashboard generation
outputs/clean/	Prepared star-schema CSV tables
outputs/dashboard.html	Interactive Python dashboard replica
PBI_Kit/	Power Query, DAX, theme and Power BI build guide
README.md	Setup and execution instructions